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CSES Problem Set

Sliding Window Sum

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the sum of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

The next line contains four integers x , a , b and c : the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window sums.

Constraints

- $1 \leq k \leq n \leq 10^7$
- $0 \leq x, a, b \leq 10^9$
- $1 \leq c \leq 10^9$

Example

Input:

```
8 5
3 7 1 11
```

Output:

```
12
```

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their sums are 14, 15, 22 and 27. Thus, the answer is $14 \oplus 15 \oplus 22 \oplus 27 = 12$.

CSES Problem Set

Sliding Window Minimum

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the minimum of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

The next line contains four integers x , a , b and c : the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window minimums.

Constraints

- $1 \leq k \leq n \leq 10^7$
- $0 \leq x, a, b \leq 10^9$
- $1 \leq c \leq 10^9$

Example

Input:

```
8 5
3 7 1 11
```

Output:

```
3
```

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their minimums are 0, 0, 1 and 2. Thus, the answer is $0 \oplus 0 \oplus 1 \oplus 2 = 3$.

CSES Problem Set

Sliding Window Xor

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the bitwise xor of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

The next line contains four integers x , a , b and c : the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window xors.

Constraints

- $1 \leq k \leq n \leq 10^7$
- $0 \leq x, a, b \leq 10^9$
- $1 \leq c \leq 10^9$

Example

Input:

```
8 5
3 7 1 11
```

Output:

```
0
```

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their xors are 8, 15, 8 and 15. Thus, the answer is $8 \oplus 15 \oplus 8 \oplus 15 = 0$.

CSES Problem Set**Sliding Window Or**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the bitwise or of each window of k elements, from left to right.

In this problem the input data is large and it is created using a generator.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

The next line contains four integers x , a , b and c : the input generator parameters. The input is generated as follows:

- $x_1 = x$
- $x_i = (ax_{i-1} + b) \bmod c$ for $i = 2, 3, \dots, n$

Output

Print the xor of all window ors.

Constraints

- $1 \leq k \leq n \leq 10^7$
- $0 \leq x, a, b \leq 10^9$
- $1 \leq c \leq 10^9$

Example

Input:

```
8 5
3 7 1 11
```

Output:

```
4
```

Explanation: The input array is $[3, 0, 1, 8, 2, 4, 7, 6]$. The windows are $[3, 0, 1, 8, 2]$, $[0, 1, 8, 2, 4]$, $[1, 8, 2, 4, 7]$ and $[8, 2, 4, 7, 6]$, and their ors are 11, 15, 15 and 15. Thus, the answer is $11 \oplus 15 \oplus 15 \oplus 15 = 4$.

CSES Problem Set**Sliding Window Distinct Values**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the number of distinct values in each window of k elements, from left to right.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the numbers of distinct values.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
8 3
1 2 3 2 5 2 2 2
```

Output:

```
3 2 3 2 2 1
```

CSES Problem Set**Sliding Window Mode**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the mode each window of k elements, from left to right.

The mode is the most frequent element in an array. If there are several possible modes, choose the smallest of them.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the modes.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
8 3
1 2 3 2 5 2 4 4
```

Output:

```
1 2 2 2 2 4
```

CSES Problem Set

Sliding Window Mex

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the mex of each window of k elements, from left to right.

The mex is the smallest nonnegative integer that does not appear in the array. For example, the mex for $[3, 1, 4, 3, 0, 5]$ is 2.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the mex values.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $0 \leq x_i \leq 10^9$

Example

Input:

```
8 3
1 2 1 0 5 1 1 0
```

Output:

```
0 3 2 2 0 2
```

CSES Problem Set**Sliding Window Median**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the median of each window of k elements, from left to right.

The median is the middle element when the elements are sorted. If the number of elements is even, there are two possible medians and we assume that the median is the smaller of them.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the medians.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
8 3
2 4 3 5 8 1 2 1
```

Output:

```
3 4 5 5 2 1
```

CSES Problem Set

Sliding Window Cost

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate for each window of k elements, from left to right, the minimum total cost of making all elements equal.

You can increase or decrease each element with cost x where x is the difference between the new and the original value. The total cost is the sum of such costs.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Output $n - k + 1$ values: the costs.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
8 3
2 4 3 5 8 1 2 1
```

Output:

```
2 2 5 7 7 1
```

CSES Problem Set**Sliding Window Inversions**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the number of inversions in each window of k elements, from left to right.

An inversion is a pair of elements where the left element is larger than the right element.

Input

The first line contains two integers n and k : the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n - k + 1$ values: the numbers of inversions.

Constraints

- $1 \leq k \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input:

```
8 3
1 2 3 2 5 2 4 4
```

Output:

```
0 1 1 1 2 0
```